

Maharashtra Public Service Commission

Assessment

Question Paper Name :	Deputy Engineer Mechanical 05th April 2025 Shift 2
Subject Name :	Deputy Engineer Mechanical
Creation Date :	2025-04-05 15:25:48
Duration :	60
Total Marks :	200
Display Marks:	Yes
Share Answer Key With Delivery Engine :	Yes
Actual Answer Key :	Yes

Deputy Engineer Mechanical

Group Number :	1
Group Id :	5623034
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	200

Deputy Engineer Mechanical

Section Id :	5623034
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100
Number of Questions to be attempted :	100
Section Marks :	200
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	5623037
Question Shuffling Allowed :	Yes

Question Number : 1 Question Id : 562303301 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Who is credited with pioneering work in microwave, optics, which became a corner stone of wireless technology ?

Options :

1. Homi Bhabha
2. Jagdish Chandra Bose
3. Vikram Sarabhai
4. M. N. Saha

Question Number : 2 Question Id : 562303302 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which field did Janaki Ammal's genetic research impact the most helping boost agricultural yields ?

Options :

1. Wheat and corn
2. Sugarcane and grain
3. Cotton and soyabeans
4. Vegetables

Question Number : 3 Question Id : 562303303 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following is an achievement of biotechnology in the agriculture ?

Options :

1. Decreasing crop production
2. Increasing resistance to diseases and drought

3. Reducing the nutritional value of food products
4. Increasing dependence on insecticides

Question Number : 4 Question Id : 562303304 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is the need for conducting an energy audit ?

Options :

1. To increase energy consumption
2. To assess environmental impacts only
3. To identify and reduce energy wastage
4. To promote energy efficiency technologies

Question Number : 5 Question Id : 562303305 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

How has development thinking of outsiders evolved regarding rural development ?

Options :

1. Initially focused on growth and modernisation
2. Always prioritized socialist forms of organisation
3. Never considered services for basic needs
4. Ignore the desires of rural development communities

Question Number : 6 Question Id : 562303306 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following reasons professionals are drawn to urban areas rather than rural areas ?

Options :

1. Lower salaries and career prospectus
2. Limited access to consumer goods
3. Better house, hospitals and schools
4. Scarce communication facilities

Question Number : 7 Question Id : 562303307 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is the main advantage of using Genetically Modified (GM) technology in agriculture ?

Options :

1. Increased need for insecticide sprays
2. Decreased crop productivity
3. Increased productivity through new varieties with pest resistance
4. Higher dependence on chemical fertilizers

Question Number : 8 Question Id : 562303308 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which Indian leader made a notable plea for compulsory primary education in the imperial legislative council, though his proposal was not accepted ?

Options :

1. Jawaharlal Nehru
2. Sardar Patel
3. Gopal Krishna Gokhale

4. B. R. Ambedkar

Question Number : 9 Question Id : 562303309 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which Amendment and Mission are mentioned as India's first step towards Urban reforms ?

Options :

1. 73rd Amendment and Smart Cities Mission
2. 74th Amendment and Jawaharlal Nehru National Urban Renewal Mission (JNNURM)
3. 75th Amendment and Housing for All Mission
4. 76th Amendment and AMRUT Mission

Question Number : 10 Question Id : 562303310 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

How does sociologist typically view towards modernization ?

Options :

1. As socio-economic change
2. As historical process
3. As a social change from traditional to modern society
4. As a technological advancement

Question Number : 11 Question Id : 562303311 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In the multiple disc clutch, if there are 8 discs on the driven shaft and 9 discs on the driving shaft then the number of pairs of contact surfaces will be

Options :

1. 16
2. 64
3. 72
4. 81

Question Number : 12 Question Id : 562303312 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

A _____ can be defined as a mechanical device that permanently joins two rotating shafts to each other.

Options :

1. Splines
2. Coupling
3. Clutch
4. Key

Question Number : 13 Question Id : 562303313 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

_____ is one of the most common modes of failure in engines and rotating shafts when applied loads are cyclic.

Options :

1. Fracture
2. Fatigue
3. Corrosion
4. Deflection

Question Number : 14 Question Id : 562303314 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The role of universal joints and _____ is to transmit torques between shafts which are not in line.

Options :

1. Crank shafts
2. Cam shafts
3. Drive shafts
4. Connecting rods

Question Number : 15 Question Id : 562303315 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Read the following :

The coefficient of friction between the belt and the pulley depends upon the following factors

- a. The material of the belt
- b. The arc of contact between belt and pulley
- c. The slip of belt
- d. The speed of belt

Which of the above factor is not correct ?

Options :

1. d
2. b
3. c
4. a

Question Number : 16 Question Id : 562303316 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

For minimal frictional loss in the screen opening and rate of incrustation and corrosion, the screen entrance velocity

Options :

1. is equal to or less than 3 cm per second
2. is equal to or more than 3 cm per second
3. is equal to or more than 3 m per second
4. is equal to or less than 3 m per second

Question Number : 17 Question Id : 562303317 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

If the well screen entrance velocity is _____ per second, friction losses in the screen openings will be negligible.

Options :

1. Equal to or less than 7 cm
2. Equal to or less than 3 cm
3. Greater than 7 cm
4. Greater than 5 cm

Question Number : 18 Question Id : 562303318 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the name of the imaginary inverted cone formed around a well due to water pumping ?

Options :

1. Cone of influence
2. Cone of depression

3. Circle of influence
4. Radius of depression

Question Number : 19 Question Id : 562303319 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

_____ is the distance by which the water level in a well is lowered from static level when pumped at a constant rate of flow.

Options :

1. Draw up
2. Draw down
3. Over pumping
4. Water length

Question Number : 20 Question Id : 562303320 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following, for boundary of the area affected by the pumping of a well ?

Options :

1. Circle of depression
2. Circle of influence
3. Cone of influence
4. Radius of influence

Question Number : 21 Question Id : 562303321 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Drilling with _____ combines the percussive action of percussive drilling and rotary action of rotary drilling.

Options :

1. Down-the-hole
2. Up-the-hole
3. Hole-the down
4. Hole-the-up

Question Number : 22 Question Id : 562303322 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The design elements in tube wells include :

- i) The well diameter
- ii) Well depth and screen
- iii) Gravel pack
- iv) Well spacing

Options :

1. (i) and (iv) only
2. (ii), (iii) and (iv) only
3. (i), (ii) and (iv) only
4. All of the above

Question Number : 23 Question Id : 562303323 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The properties of drilling mud that influences its performance.

Options :

1. mud viscosity and density

2. gel strength
3. filtration property and sand content
4. all of the above

Question Number : 24 Question Id : 562303324 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Specific yield (S_y) of the aquifer is given by

Options :

1.
$$S_y = \frac{\text{Volume of water drained}}{\text{Total of volume of the saturated rock or soil}} \times 100$$
2.
$$S_y = \frac{\text{Total of volume of the saturated rock or soil}}{\text{Volume of water drained}} \times 100$$
3.
$$S_y = \frac{\text{Volume of water drained}}{\text{Volume of water imposed}} \times 100$$
4. None of the above

Question Number : 25 Question Id : 562303325 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

_____ is the upper most section of the drill stem.

Options :

1. Sub
2. Kelly
3. Drill collar
4. Drill pipe

Question Number : 26 Question Id : 562303326 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which among the following are the desired characteristics of efficient cooling system ?

Options :

1. It should be capable of removing about 30% heat generated in the combustion chamber while maintaining the optimum temperature of the engine
2. It should remove heat at faster rate when engine is hot. However during starting of engine the cooling should be minimum
3. Both above (1) and (2)
4. None of the above

Question Number : 27 Question Id : 562303327 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

If the power requirements in application such as automobile can be met with either a naturally aspirated SI engine of certain size or with a smaller size engine which is turbocharged to the same maximum power, which engine shall offer better fuel economy at part load ?

Options :

1. Naturally aspirated S.I. engine
2. Turbocharged engine
3. Both engines will offer same fuel economy
4. Cannot predicted for better fuel economy

Question Number : 28 Question Id : 562303328 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In I.C. Engines, boundary lubrication may occur

- A. When the piston and piston rings are at the beginning and end of stroke.
- B. Between the piston pin and bushing.
- C. Between the gear teeth of these statements.

Options :

- 1. Only (A) is correct
- 2. (A) and (B) are correct
- 3. (A), (B) and (C) are correct
- 4. Only (C) is correct

Question Number : 29 Question Id : 562303329 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The main purpose of a thermostat in an engine cooling system is to

Options :

- 1. allow engine to warm-up quickly
- 2. prevent the coolant from boiling
- 3. pressurize the system
- 4. indicate to the driver the coolant temperature

Question Number : 30 Question Id : 562303330 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The purpose of the inertia switch in the in-tank fuel-pumping wiring circuit is to

Options :

- 1. maintain the inertia of the fuel pump to prevent fuel starvation of the engine
- 2. shut off the fuel pump if the car rolls over

3. keep inertia from stalling the engine
4. prevent inertia from shutting off the fuel pump

Question Number : 31 Question Id : 562303331 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In case of centrifugal pump, the range of water velocity in delivery pipe is

Options :

1. 3.5 m/sec to 4.1 m/sec
2. 4.5 m/sec to 5.2 m/sec
3. 1.5 m/sec to 3.5 m/sec
4. 3 m/sec to 6.0 m/sec

Question Number : 32 Question Id : 562303332 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

When there is sudden power failure, the kinetic energy of the rotating parts becomes small and it cannot maintain flow against the discharge head ? The pump starts immediately behaving like turbine, this phenomenon is called as

Options :

1. Priming
2. Cavitation
3. Hammering
4. None of the above

Question Number : 33 Question Id : 562303333 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In axial flow impeller, the maximum distance from chord to meanline is known as

Options :

1. Camber
2. Caster
3. Incidence angle
4. None of the above

Question Number : 34 Question Id : 562303334 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which among the following is not the positive displacement pump ?

Options :

1. Gear pump
2. Screw pump
3. Vane pump
4. Volute type centrifugal pump

Question Number : 35 Question Id : 562303335 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

A vane pump is to have a volumetric displacement of 75 cm^3 . It has a rotor diameter of 5 cm, a cam ring diameter of 10 cm and a vane width of 5 cm. What must be the eccentricity ?

Options :

1. $\frac{2}{\pi} \text{ cm}$
2. $\pi/2 \text{ cm}$

3. 2π cm

4. $\frac{3}{\pi}$ cm

Question Number : 36 Question Id : 562303336 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is manometric head in centrifugal pump ?

Options :

1. Vertical distance from center line of the pump to the free surface in the tank to be pumped
2. Vertical distance from center line of the pump to the free surface of liquid in well
3. It is static suction head plus velocity head
4. It is total head that must be produced by the pump

Question Number : 37 Question Id : 562303337 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is Net Positive Suction Head (NPSH) in pumps ?

Options :

1. It is total suction head in meter absolute determined at the inlet nozzle of the pump and corrected to pump datum less the vapour pressure head of the liquid
2. It is manometric head minus delivery head
3. It is total suction head minus friction head in the pipe
4. None of the above

Question Number : 38 Question Id : 562303338 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the minimum speed required to start a centrifugal pump ?

Options :

1. The speed required to achieve a pressure rise in the impeller greater than or equal to the manometric head
2. A speed that minimizes the tangential velocity of the impeller at the outlet
3. A speed that maximizes the tangential velocity of the impeller at the inlet
4. The speed required to achieve a flow rate equal to the design flow rate

Question Number : 39 Question Id : 562303339 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

In case of indicator diagram of reciprocating pump, what does the area enclosed by the loop in an indicator diagram represent ?

Options :

1. The total volume of fluid displaced by the pump per cycle
2. The net work done per cycle by the pump on the fluid
3. The pressure difference between the suction and delivery sides of the pump
4. The efficiency of the pump during a single cycle

Question Number : 40 Question Id : 562303340 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the main purpose of the air vessel in a piping system ?

Options :

1. To regulate water pressure
2. To remove air bubbles from the water

3. To block the flow of water in the pipe
4. To measure the flow rate of the water

Question Number : 41 Question Id : 562303341 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The whole electrical submersible pump motor is filled with a highly refined oil because it provides

Options :

1. The required dielectric strength to prevent short circuits between motor parts
2. The proper lubrication of the bearing
3. The good thermal conductivity required to carry the heat generated in the motor to the outside of the housing
4. All of the above

Question Number : 42 Question Id : 562303342 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In case of a D.C. series motor, due to increase in the conduction angle of thyristor, power delivered to the motor is increased with a subsequent increase in its

Options :

1. load
2. speed
3. friction loss
4. heat generation

Question Number : 43 Question Id : 562303343 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In steady state operation of D.C. motor, current will be determined by the

Options :

1. mechanical load torque
2. speed of the motor
3. voltage supplied
4. mechanical and electrical load

Question Number : 44 Question Id : 562303344 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What special adaptation is included in heavy-duty submersible pumps for pumping sludge and heavy muds ?

Options :

1. Rubber coated impellers
2. Multistage impellers
3. Oil cooled motor
4. Agitator mounted on the shaft end

Question Number : 45 Question Id : 562303345 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The electrical submersible pump motors are most reliable due to the fact, their rotor

Options :

1. is not connected to the electric supply
2. is directly connected to the electric supply
3. rotates at very high speed

4. rotates at very low speed

Question Number : 46 Question Id : 562303346 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In KT double bridge method, as per BIS, in the test of 3-phase, induction motors, the variation of resistance between phases is permitted to the extent of

Options :

1. 3%
2. 5%
3. 7%
4. 3.5% to 4.5%

Question Number : 47 Question Id : 562303347 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is one of the advantage of DC motors over AC motors in pumps applications ?

Options :

1. They can be operated without any maintenance
2. They have synchronous speeds
3. They have faster response to transient conditions
4. They are more efficient at constant speed

Question Number : 48 Question Id : 562303348 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

At what Miller number should wear problems be considered ?

Options :

1. less than 50

2. between 25 and 50
3. between 10 and 40
4. more than 50

Question Number : 49 Question Id : 562303349 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following safety measures is used in heavy duty submersible pumps to protect the motor in case of overheating ?

Options :

1. Single phase motor protection
2. Rubber coated motor windings
3. Built in thermal protection unit with thermistors
4. External cooling fans

Question Number : 50 Question Id : 562303350 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

How is the speed of a DC motor controlled upto its base speed ?

Options :

1. By varying the frequency of the power supply
2. By adjusting the armature voltage
3. By controlling the torque output
4. By altering the mechanical load

Question Number : 51 Question Id : 562303351 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The effectiveness of the inter-cooler used in reciprocating compressor is given by which of the following ?

Options :

1.
$$\frac{\text{Actual drop in temperature of air}}{\text{Maximum possible drop in temperature of air}}$$
2.
$$\frac{\text{Actual drop in temperature of air}}{\text{Theoretical drop in temperature of air}}$$
3.
$$\frac{\text{Actual heat transfer}}{\text{Maximum possible heat transfer}}$$
4. None of the above

Question Number : 52 Question Id : 562303352 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which among the following is the correct assumption used for minimum work required to compress air in multi-stage compressor ?

Options :

1. The air is cooled to initial temperature after each stage of compressor
2. The pressure ratio in each stage is different
3. Work done in all the stages is different, if the same index of compression is used
4. All the above

Question Number : 53 Question Id : 562303353 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What makes screw compressors more efficient compared to reciprocating compressor ?

Options :

1. Lower rotational speed
2. Ability to handle higher pressure
3. Higher flow rate
4. No clearance volume

Question Number : 54 Question Id : 562303354 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which control method involves restricting gas flow into the compressor inlet to regulate flow and pressure ?

Options :

1. Recycle valve control
2. Suction valve throttling
3. Variable guide vane control
4. Discharge valve throttling

Question Number : 55 Question Id : 562303355 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following control techniques is unique for screw compressors ?

Options :

1. Speed control
2. Discharge valve throttling
3. Slide valve
4. Suction valve throttling

Question Number : 56 Question Id : 562303356 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In reciprocating compressor control, in the "compressor-auto-mode" position why the Prelube pump is set to run continuously ?

Options :

1. To maintain oil temperature
2. To maintain oil pressure
3. (1) and (2) as above
4. None of the above

Question Number : 57 Question Id : 562303357 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The partial breakdown of air flow in a Rotary compressor often leads to more severe instability is called as

Options :

1. Surge
2. Stall
3. Breakdown
4. Drag

Question Number : 58 Question Id : 562303358 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Where is the Capacity gauge used as an estimator during modulation controls ?

Options :

1. In Centrifugal compressors

2. In Reciprocating compressors
3. In Rotary screw compressors
4. None of the above

Question Number : 59 Question Id : 562303359 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The slip factor in a Rotary compressor is defined as the ratio of

Options :

1. Ideal whirl component to actual whirl component
2. Actual whirl component to ideal whirl component
3. Isentropic work to actual work
4. None of the above

Question Number : 60 Question Id : 562303360 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In Reciprocating air compressor, which of the following statement is correct ?

Options :

1. Work of compression with clearance is more as compared to without clearance
2. Work of compression without clearance is more as compared to with clearance
3. Clearance does not affect the work of compression per kg of air
4. None of the above

Question Number : 61 Question Id : 562303361 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What are two inter related aspects that must be considered when deciding an engine layout ?

Options :

1. Engine size and weight
2. Engine balance and firing interval between the cylinders
3. Engine cost and efficiency
4. Engine power and torque

Question Number : 62 Question Id : 562303362 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Weight of the vehicle along with the passengers produces which of the action in the side members of the chassis frame ?

Options :

1. Lateral bending
2. Torsion
3. Horizontal bending
4. Vertical bending

Question Number : 63 Question Id : 562303363 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

_____ is the basic need of vehicle design that decides the target market and brand image of the vehicle. It also decides the locations of the engine and transmission system.

Options :

1. Body design
2. Vehicle layout

3. Wheel drive layout
4. Transmission assembly

Question Number : 64 Question Id : 562303364 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

How does a chassis dynamometer help in diagnosing vehicle problems ?

Options :

1. By reducing engine noise
2. By simplifying the engine design
3. By allowing test equipment to be connected and monitored while the vehicle is stationary
4. By improving vehicle aerodynamics

Question Number : 65 Question Id : 562303365 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What does the product cost dimension include ?

Options :

1. The time taken to develop the product
2. The overall price customer are willing to pay
3. Spending on capital equipment, tooling and incremental cost of producing each unit
4. The marketing and distribution cost

Question Number : 66 Question Id : 562303366 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

A valve with an ideal linear inherent flow characteristic, throughout the travel range, produces flow rate directly proportional to the amount of

Options :

1. Valve inner diameter
2. Valve outer diameter
3. Valve plug travel
4. Valve alignment with seat ring

Question Number : 67 Question Id : 562303367 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In vapour compression refrigeration systems, the major operating cost is the energy input to the system in the form of

Options :

1. Electrical load
2. Mechanical work
3. Volume of refrigerant
4. All of the above

Question Number : 68 Question Id : 562303368 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Accumulators are devices which are used

Options :

1. As auxiliary power source
2. To serve as leakage compensators
3. To serve as shock absorber

4. All of the above

Question Number : 69 Question Id : 562303369 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In case of non return valves, a check valve with a heavy spring may act as a

Options :

1. Pressure control valve
2. Pressure relief valve
3. Oil reservoir
4. None of the above

Question Number : 70 Question Id : 562303370 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

For a hydraulic circuit selection of pump depends upon

Options :

1. Maximum pressure and maximum flow
2. Maximum temperature and maximum volume
3. Maximum pressure and maximum temperature
4. Minimum pressure and maximum flow

Question Number : 71 Question Id : 562303371 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which one of the following is the mechanical transducer ?

Options :

1. Amplifier

2. Oscillator
3. Filter
4. Dial gauge

Question Number : 72 Question Id : 562303372 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

_____ is the popular non contracting rotary speed transducer and is known as variable reluctance tachometer generator.

Options :

1. Linear velocity transducer
2. Linear hall effect transducer
3. Pulse speed transducer
4. Encoders

Question Number : 73 Question Id : 562303373 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the purpose of regenerative circuit in the hydraulic cylinder ?

Options :

1. It is used to speed up the retracting speed of the double acting hydraulic cylinder
2. It is used to unload the pump
3. It is used to speed up the extending speed of the double acting hydraulic cylinder
4. None of the above

Question Number : 74 Question Id : 562303374 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Single acting cylinders are not preferred in the applications of larger stroke lengths due to

Options :

1. Oil viscosity problem
2. Manufacturing cost problem
3. Return spring mechanical property problem
4. Mounting problem

Question Number : 75 Question Id : 562303375 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The complete valve description needs

Options :

1. Number of ports and number of position
2. Number of ports and the control action
3. Number of ports, number of position and the control action
4. Number of position and the control action

Question Number : 76 Question Id : 562303376 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which of the following transducers is passive transducer ?

Options :

1. Photovoltaic cell
2. Piezoelectric
3. Thermocouple

4. None of the above

Question Number : 77 Question Id : 562303377 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Match the Parts 'A' and 'B' in connection with gauges :

Part – A

Part – B

- | | |
|--|----------------|
| i. The basic industrial standards used as reference and setting means for comparative measurements in precision manufacturing. | a. Ring gauge |
| ii. A gauge outside measuring surfaces of which are designed to verify the specified uniformity of holes. | b. Plug gauge |
| iii. A gauge the inside measuring surfaces of which are circular in form. | c. Gauge block |
| iv. A fixed gauge arranged with inside measuring surfaces for calipering diameters, length, thickness or width. | d. Snap gauge |

Options :

1. i-c, ii-a, iii-d, iv-b
2. i-c, ii-b, iii-a, iv-d
3. i-a, ii-b, iii-c, iv-d
4. i-c, ii-b, iii-d, iv-a

Question Number : 78 Question Id : 562303378 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In case of solar pump controller MPPT stands for

Options :

1. Motor Power Point Technique
2. Maximum Power Point Tracker

3. Minimum Power Point Transfer
4. Maximum Power Penetration Technique

Question Number : 79 Question Id : 562303379 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

High voltage test (at room temperature) multi core cables and cords shall withstand without breakdown an ac voltage of _____ KV (rms) or a dc voltage of _____ KV for a period of 5 min for each test connection.

Options :

1. For AC – 4 KV or DC – 6.5 KV
2. For AC – 5 KV or DC – 7 KV
3. For AC – 3 KV or DC – 7.2 KV
4. For AC – 3.5 KV or DC – 7.2 KV

Question Number : 80 Question Id : 562303380 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

As per Indian standard 4759, for fabricated steel structure having thickness 5 mm or above, the minimum/average mass of zinc coating (g/m^2) in the process of Hot dip galvanising is

Options :

1. $610 \text{ (g/m}^2\text{)}$
2. $460 \text{ (g/m}^2\text{)}$
3. $510 \text{ (g/m}^2\text{)}$
4. $340 \text{ (g/m}^2\text{)}$

Question Number : 81 Question Id : 562303381 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In solar cell, adding which of the element into a pure germanium, an n-type semiconductor is formed ?

Options :

1. Arsenic impurity
2. Aluminium
3. Magnesium
4. Free electron

Question Number : 82 Question Id : 562303382 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

In solar PV panel, when modules are connected in series, at what same parameter it is desirable that each module should produce maximum power ?

Options :

1. Same current
2. Same voltage
3. Same resistance
4. Same modules

Question Number : 83 Question Id : 562303383 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

To design future storage systems, scientists and engineers need to explore new media for thermal energy storage, mainly, ceramics, concrete and test alternative thermochemical cycles. Adding nanoparticles to increase the heat capacity of _____ is an option.

Options :

1. Silicon
2. Molten salts
3. Lithium-ion batteries
4. Water

Question Number : 84 Question Id : 562303384 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

The efficiency of the conversion of light to electricity in photovoltaic cells usually falls with _____ because after thermal activation, the generation of photons is no longer so possible.

Options :

1. constant temperature
2. decreasing temperature
3. increasing temperature
4. none of the above

Question Number : 85 Question Id : 562303385 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Solar cells are solid-state devices that convert sunlight directly into electricity, without the need for any mechanical or moving parts and are made from thin slice or layer of _____ materials.

Options :

1. Conductor
2. Semiconductor
3. Non-conductor

4. Super conductor

Question Number : 86 Question Id : 562303386 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

_____ is the thermochemical conversion technologies that use solid biomass as a primary fuel for the production of electricity. In addition to this, the use of liquid biomass (such as sewage sludge) for the production of methane via anaerobic digestion is increasingly common.

Options :

1. Chemical burning
2. Pyrolysis
3. Carbonization
4. None of the above

Question Number : 87 Question Id : 562303387 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What materials are mentioned as being used directly into solar modules in the form of thin film ?

Options :

1. Amorphous silicon and cadmium telluride
2. Crystalline silicon and gallium arsenide
3. Copper indium gallium selenide and perovskite
4. Polycrystalline silicon and amorphous silicon

Question Number : 88 Question Id : 562303388 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is the key characteristics of polar tracking in solar concentrator system ?

Options :

1. The concentrator rotates about a horizontal axis
2. The concentrator rotates about an axis parallel to the earth's axis of rotation
3. The panels are fixed and do not rotate
4. The concentrator follows the movement of the moon

Question Number : 89 Question Id : 562303389 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Which of the following is not a common type of solar panel ?

Options :

1. Monocrystalline
2. Polycrystalline
3. Amorphous silicon
4. Carbon fiber

Question Number : 90 Question Id : 562303390 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What happens to the electrons released during the oxidation reaction in a PEM, fuel cell ? (Proton Exchange Membrane fuel cell)

Options :

1. They combine with hydrogen ions to form water
2. They pass through an external circuit to generate an electric current
3. They react with oxygen to form carbon dioxide
4. They remain at the anode

Question Number : 91 Question Id : 562303391 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

What is the primary function of a Capacitor in terms of energy storage ?

Options :

1. To convert mechanical energy into electrical energy
2. To reversibly store energy in electric fields and interfaces
3. To produce continuous electrical current
4. To generate thermal energy

Question Number : 92 Question Id : 562303392 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Solar cells do not operate at theoretical maximum efficiency because of

Options :

1. reduction losses at the top surfaces and fill factor loss
2. incomplete combustion of fuel
3. heat loss to the environment
4. diffusion of energy

Question Number : 93 Question Id : 562303393 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Oceans serve as a tremendous store house of solar energy, because of the _____ produced by the sun as well as the _____ energy stored in the waves.

Options :

1. pressure difference, kinetic

2. heat, potential
3. temperature difference, kinetic
4. pressure, potential

Question Number : 94 Question Id : 562303394 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Wind machines are generally classified in terms of orientation of the axis of rotation of their rotors as _____ and _____ machines.

Options :

1. horizontal, vertical
2. inclined, stator
3. universal, non universal
4. step-up, step-down

Question Number : 95 Question Id : 562303395 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Solar cells convert the solar radiation directly into electricity using _____ effect without going through a thermal process.

Options :

1. photosynthesis
2. photovoltaic
3. photoelectric
4. mechanical

Question Number : 96 Question Id : 562303396 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Calculate "Fill Factor" using following data :

$$P_{\max} = 32 \text{ W}, V_{\text{oc}} = 8 \text{ V}, I_{\text{sc}} = 4 \text{ A}$$

Options :

1. 64
2. 16
3. 1
4. 5

Question Number : 97 Question Id : 562303397 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

If no load is connected to a solar PV system

Options :

1. it will stop absorbing light
2. it will dissipate energy in the panel and increase its temperature
3. its voltage will go on increasing till its breakdown
4. it will start reflecting the light

Question Number : 98 Question Id : 562303398 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The absorber plate which is used in liquid flat plate solar collectors, made from a metal sheet has a thickness range of

Options :

1. 0.2 to 0.7 mm
2. 0.5 to 1.5 mm
3. 1.5 to 3.5 mm

4. 3.5 to 7.0 mm

Question Number : 99 Question Id : 562303399 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Use of booster mirrors with a flat plate collector

Options :

1. Increases the reflection to the atmosphere
2. Decreases the reflection to the atmosphere
3. Increases the diffused radiation on the absorber
4. Increases the beam radiation on the absorber

Question Number : 100 Question Id : 562303400 Question Type : MCQ Option Shuffling : No Correct Marks : 2 Wrong Marks : 0.5

Flat plate collector is one of the most important type of solar collector which can be used for variety of applications in which the required temperature range is,

Options :

1. 5°C to 20°C
2. 20°C to 60°C
3. 40°C to 100°C
4. 30°C to 50°C